

Mathematical Description of the Bubble Size Histogram Pipeline

1. Setup and Notation

Let $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$. A grayscale image is a function $I : \Omega \rightarrow [0, 1]$ where $\Omega = \{0, \dots, H-1\} \times \{0, \dots, W-1\} \subset \mathbb{Z}^2$ is the pixel lattice. Images are stored as $H \times W$ arrays of type `float32`.

A **bubble annotation** is a triple $(c, r) \in \mathbb{R}^2 \times \mathbb{R}_{>0}$ specifying a center $c = (c_x, c_y)$ and radius r . The annotated training corpus is

$$\mathcal{D} = \{(I_k, \mathcal{B}_k)\}_{k=1}^N, \quad \mathcal{B}_k = \{(c_{k,j}, r_{k,j})\}_{j=1}^{M_k},$$

where $N = 14$ images and $\sum_k M_k = 5,584$ annotated bubbles (radii ranging from 0.81 to 263.6 px; quartiles $Q_1 = 4.2$, $Q_2 = 7.0$, $Q_3 = 10.3$ px).

Polygon annotations. Labelling encodes some bubbles as polygons $\{v_i\}_{i=1}^n \subset \mathbb{R}^2$. These are converted to the (c, r) form via the minimum enclosing circle heuristic:

$$c = \frac{1}{n} \sum_{i=1}^n v_i, \quad r = \max_i \|v_i - c\|_2.$$

Hyperparameters. All hyperparameters are collected in a single configuration object:

Symbol	Parameter	Default
s	<code>template_size</code>	10
σ	<code>scale_factor</code>	0.9
$r_{\min}, r_{\max}$	<code>min_radius, max_radius</code>	1.0, 50.0
K	<code>num_templates</code>	1
n_b	<code>n_score_bins</code>	50
η	<code>neg_sample_ratio</code>	10
d	<code>min_neg_dist</code>	10

2. Data Split

The N images are partitioned into three disjoint index sets $\mathcal{I}_T, \mathcal{I}_C, \mathcal{I}_{\text{test}}$ by applying a seeded random permutation π of $\{1, \dots, N\}$:

$$n_{\text{test}} = \max(1, \text{round}(N(1 - f_T - f_C))), \quad n_T = \text{round}(Nf_T),$$

$$\begin{aligned} \mathcal{I}_{\text{test}} &= \{\pi(1), \dots, \pi(n_{\text{test}})\}, \quad \mathcal{I}_T = \{\pi(n_{\text{test}} + 1), \dots, \pi(n_{\text{test}} + n_T)\}, \\ \mathcal{I}_C &= \{\pi(n_{\text{test}} + n_T + 1), \dots, \pi(N)\}. \end{aligned}$$

Default fractions: $f_T = 0.30$, $f_C = 0.65$, giving $|\mathcal{I}_T| = 4$, $|\mathcal{I}_C| = 9$, $|\mathcal{I}_{\text{test}}| = 1$ for $N = 14$.

Images in $\mathcal{I}_T$ are used exclusively for template construction; images in $\mathcal{I}_C$ exclusively for calibration. The sets are never mixed. Images in $\mathcal{I}_{\text{test}}$ are held out entirely.

3. Template Construction

3.1 Patch Extraction

For each annotated bubble (c, r) in image I_k with $k \in \mathcal{I}_T$, let $\hat{r} = \max(1, \text{round}(r))$. Extract the square patch

$$P = I_k[\lfloor c_y \rfloor - \hat{r} : \lfloor c_y \rfloor + \hat{r}, \lfloor c_x \rfloor - \hat{r} : \lfloor c_x \rfloor + \hat{r}] \in [0, 1]^{2\hat{r} \times 2\hat{r}},$$

discarding any bubble whose bounding box extends outside Ω .

3.2 Normalisation and Resizing

Apply a bilinear resize with anti-aliasing to obtain $\tilde{P} \in \mathbb{R}^{s \times s}$. Normalise to unit ℓ^1 mass:

$$\hat{P} = \frac{\tilde{P}}{\|\tilde{P}\|_1},$$

so that every patch contributes equally regardless of absolute brightness. Patches with $\|\tilde{P}\|_1 = 0$ are discarded.

3.3 Size Bins and Averaging

The radius range $[r_{\min}, r_{\max}]$ is partitioned into K log-spaced bins with edges

$$e_k = r_{\min} \left(\frac{r_{\max}}{r_{\min}} \right)^{k/K}, \quad k = 0, \dots, K,$$

and geometric-mean centres $\mu_k = \sqrt{e_k e_{k+1}}$. Each bubble is assigned to bin $\kappa(r) = \min(\lfloor \log_{r_{\max}/r_{\min}}(r/r_{\min}) \cdot K \rfloor, K - 1)$.

Let $\mathcal{P}_k$ denote the set of normalised patches assigned to bin k . The raw template for bin k is the arithmetic mean:

$$\bar{T}_k = \frac{1}{|\mathcal{P}_k|} \sum_{\hat{P} \in \mathcal{P}_k} \hat{P} \in \mathbb{R}^{s \times s}.$$

The final template is ℓ^2 -normalised:

$$T_k = \frac{\bar{T}_k}{\|\bar{T}_k\|_2} \in \mathbb{R}^{s \times s}, \quad \|T_k\|_2 = 1.$$

The ℓ^2 normalisation is theoretically motivated by Pedro’s formulation (making T_k a unit vector so the NCC equals a cosine similarity), but is redundant in practice since `skimage.match_template` normalises the template internally.

With $K = 1$ (default) all patches pool into a single template T_0 .

4. Image Pyramid

4.1 Levels and Effective Radii

Define the number of pyramid levels

$$L = \left\lceil \frac{\log(r_{\max}/(s/2))}{\log(1/\sigma)} \right\rceil.$$

With defaults ($s = 10$, $\sigma = 0.9$, $r_{\max} = 50$): $L = 22$.

At level $\ell \in \{0, \dots, L - 1\}$, the image is downsampled by

$$\alpha_\ell = \sigma^\ell,$$

yielding $I_\ell = \text{rescale}(I, \alpha_\ell) \in [0, 1]^{H_\ell \times W_\ell}$ with $H_\ell = \lfloor \alpha_\ell H \rfloor$, $W_\ell = \lfloor \alpha_\ell W \rfloor$. Level 0 is the original image ($\alpha_0 = 1$, $I_0 = I$).

The **effective radius** at level ℓ is the radius (in pixels of the original image I) of a bubble that appears with radius $s/2$ pixels in I_ℓ :

$$\rho_\ell = \frac{s/2}{\alpha_\ell} = \frac{s}{2} \sigma^{-\ell}.$$

The sequence (ρ_ℓ) is strictly increasing: $\rho_0 = s/2 = 5$ px through $\rho_{21} \approx 45.7$ px with defaults. Each pyramid level corresponds to one bin of the output size histogram.

4.2 Template Assignment

The template used at level ℓ is

$$T^{(\ell)} = T_{k^*(\ell)}, \quad k^*(\ell) = \arg \min_{k \in \{0, \dots, K-1\}} |\rho_\ell - \mu_k|.$$

With $K = 1$: $T^{(\ell)} = T_0$ for all ℓ .

5. Normalised Cross-Correlation Score Maps

For each pyramid level ℓ , `skimage.match_template` computes the score map $C_\ell : \Omega_\ell \rightarrow [-1, 1]$ defined at each position $(x, y) \in \Omega_\ell$ by

$$C_\ell(x, y) = \frac{\langle W_{x,y}, T^{(\ell)} \rangle}{\|W_{x,y}\|_2 \|T^{(\ell)}\|_2},$$

where $W_{x,y} \in \mathbb{R}^{s \times s}$ is the patch of I_ℓ centred at (x, y) (with reflected boundary padding), and $\langle \cdot, \cdot \rangle$ denotes the Frobenius inner product on $\mathbb{R}^{s \times s}$.

Since $\|T^{(\ell)}\|_2 = 1$, this simplifies to $C_\ell(x, y) = \langle W_{x,y} / \|W_{x,y}\|_2, T^{(\ell)} \rangle$, i.e. the cosine similarity between the vectorised normalised patch and the template. $C_\ell(x, y) \in [-1, 1]$ with $C_\ell(x, y) = 1$ iff $W_{x,y} \propto T^{(\ell)}$ (same direction in $\mathbb{R}^{s^2}$).

6. Bayesian Calibration

6.1 Positive Score Sampling

For each calibration image I_k with $k \in \mathcal{I}_C$, and each annotated bubble $(c, r) \in \mathcal{B}_k$:

1. Select the level best matching the annotated radius: $\ell^*(c, r) = \arg \min_\ell |\rho_\ell - r|$.
2. Map the bubble centre to scaled coordinates: $\tilde{c} = \alpha_{\ell^*} \cdot c \in \mathbb{R}^2$.
3. Record the positive score: $s_{k,j}^+ = C_{\ell^*}(\lfloor \tilde{c} \rfloor)$, provided $\lfloor \tilde{c} \rfloor \in \Omega_{\ell^*}$.

The positive score set is $\mathcal{S}^+ = \{s_{k,j}^+\}$.

6.2 Negative Score Sampling

Negatives are sampled from the level-0 score map C_0 (full resolution). For each calibration image I_k , define the exclusion set

$$\mathcal{E}_k = \{(x, y) \in \Omega_0 : \exists j, \|(x, y) - c_{k,j}\|_\infty \leq d\},$$

i.e. a square neighbourhood of half-side d around every annotated bubble centre. Candidate negative positions are $\mathcal{N}_k = \Omega_0 \setminus \mathcal{E}_k$. A random subset of size $\min(\eta \cdot |\mathcal{S}^+|, |\mathcal{N}_k|)$ is drawn uniformly without replacement from $\mathcal{N}_k$, yielding scores $\mathcal{S}^-$.

6.3 Empirical Likelihood Estimation

Partition $[-1, 1]$ into n_b equal-width bins $B_1, \dots, B_{n_b}$ of width $h = 2/n_b$. Estimate the score densities at bubble and non-bubble locations by normalised histograms:

$$\hat{f}^+(s) = \frac{|\{s_i^+ \in B_m : s \in B_m\}|}{|\mathcal{S}^+| \cdot h}, \quad \hat{f}^-(s) = \frac{|\{s_i^- \in B_m : s \in B_m\}|}{|\mathcal{S}^-| \cdot h}.$$

6.4 Prior Probability

The prior probability that a randomly chosen pixel location in a training image contains a bubble centre is estimated by

$$\pi_0 = \frac{\sum_{k \in \mathcal{I}_C} M_k}{\sum_{k \in \mathcal{I}_C} H_k W_k}.$$

6.5 Posterior via Bayes' Rule

By Bayes' theorem, the posterior probability that location (x, y) contains a bubble given NCC score s is

$$p(s) \triangleq P(\text{bubble} \mid \text{score} = s) = \frac{\hat{f}^+(s) \pi_0}{\hat{f}^+(s) \pi_0 + \hat{f}^-(s) (1 - \pi_0)},$$

with $p(s) := 0$ wherever the denominator vanishes. The function $p : [-1, 1] \rightarrow [0, 1]$ is stored as a piecewise-constant lookup table (one value per bin).

7. Per-Frame Histogram Estimation

Given a new image I , the estimated expected bubble count at scale level ℓ is

$$\hat{N}_\ell = \sum_{(x,y) \in \Omega_\ell} p(C_\ell(x,y)).$$

The output histogram is the sequence $\{(\rho_\ell, \hat{N}_\ell)\}_{\ell=0}^{L-1}$.

Probabilistic interpretation. Define binary random variables $B_{x,y}^{(\ell)} \in \{0, 1\}$ indicating whether pixel (x, y) in I_ℓ is a bubble centre, with $P(B_{x,y}^{(\ell)} = 1) = p(C_\ell(x, y))$. By linearity of expectation,

$$E \left[\sum_{(x,y)} B_{x,y}^{(\ell)} \right] = \sum_{(x,y)} p(C_\ell(x, y)) = \hat{N}_\ell,$$

regardless of the joint distribution of the $B_{x,y}^{(\ell)}$. The formula is thus an unbiased estimator of the expected number of bubble-centre pixels at scale ℓ , under the model that each pixel independently contains a bubble centre with probability $p(C_\ell(x, y))$.

8. Remarks and Limitations

Over-counting due to NCC spatial extent. The NCC response $C_\ell(\cdot, \cdot)$ is not a delta function at bubble centres — it has spatial extent of order s pixels. Consequently, a single bubble contributes elevated scores across a neighbourhood of its centre, causing $\hat{N}_\ell$ to overcount. The magnitude of the overcounting depends on $p(\cdot)$ near the bubble score: if p is concentrated near the bubble score, the overcounting factor is approximately proportional to the area of the high-response region, which scales as s^2 . In practice the prior π_0 is small enough that $p(s)$ is small everywhere except near $s = 1$, mitigating this.

Non-overlapping assumption. The probabilistic interpretation requires that bubble centres are sufficiently separated that at most one centre occupies any pixel position in I_ℓ . This is approximately satisfied for non-overlapping bubbles at the resolution of I_ℓ .

Approximation in negative sampling. Negatives are sampled only from C_0 (level 0), yet positives are sampled from multiple levels. This means $\hat{f}^-$ reflects the score distribution at the smallest scale only. For a fully consistent calibration, negatives should be stratified across all levels; the current approach is a computationally convenient approximation.

Template averaging across scales. The template T_k is an average of patches resized to $s \times s$, drawn from bubbles of all radii assigned to bin k . Patches from small bubbles ($r < s/2$, so patch size $< s \times s$) are upsampled; patches from large bubbles are downsampled. The resulting template approximates the expected appearance of a bubble at the canonical radius μ_k , but conflates appearance variation across the full radius range of the bin. Narrowing the bins (increasing K) reduces this conflation at the cost of fewer training patches per template.

Score-based calibration is monotone only in expectation. The Bayes posterior $p(s)$ need not be monotone in s — the empirical histograms $\hat{f}^+$ and $\hat{f}^-$ can cross multiple times with limited data. Isotonic regression could be applied post hoc to enforce monotonicity if desired.